RESEARCH REPORT · VITKI

Fully tested finding

In western Uppland ʀ was written with r on five times as large a share of the runestones as in the rest of the country

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ABSTRACT

Background and aim. The ʀ of Runic Swedish had a rune of its own but is written ever more often with the ordinary r-rune during the eleventh century. The question is whether the spelling clusters geographically more than chance would give, where, and whether the cluster withstands a fixed series of sensitivity analyses.

Material. 514 Viking-Age runestones in the Swedish provinces where it can be decided how ʀ is written and which are not moved according to the Scandinavian Runic-text Database (Rundata); 72 of them are lost and known from older records.

Method. Moran's I, Getis-Ord Gi∗ and Kulldorff's spatial scan statistic (999 simulations), adjusted for the 13 traits scanned; replication in two pre-defined halves; 7 sensitivity analyses; adjustment for all 48 associations the project reports.

Results. Within 30.4 kilometres of Holm in Vittinge parish, 24 of 36 stones have at least one r for ʀ (67 percent), against 13 percent in the rest of the country; p = 0.001, adjusted for the traits q = 0.013. Same direction in both halves (85 against 12 and 57 against 15 percent), every sensitivity analysis holds, and adjusted for all associations q = 0.02. Exploratory: the carver Erik wrote r on all his 12 securely attributed stones, but the cluster remains without them (11 of 23 stones).

Conclusion. Western Uppland is the area of Sweden where ʀ was most often written with r. With Åsmund the spelling followed the carver; Erik's stones all stand in or just outside the district, so for him, as for the rest of the district, the material cannot separate carver from place. The cluster remains with the carver held constant. We propose a writing tradition that began with Erik; pronunciation, as earlier research holds, is not ruled out.

Keywords: *runestones, Viking Age, yr-rune, ʀ, r, Uppland, carvers, spatial statistics*

HIGHLIGHTS

- In western Uppland, around Simtuna, Vårfrukyrka and Tillinge, the old sound ʀ was written with the ordinary r-rune on far more runestones than anywhere else in Sweden.
- Åsmund kept the yr-rune even inside the district, Erik wrote r on all his stones, which stand in or near it, and the district carried on after Erik; whether that was the carvers' habit or the district's speech the stones cannot decide.
- Our hypothesis is a writing tradition. Earlier research has read it as pronunciation, and the stones cannot yet decide who is right.

1. Introduction

The ʀ of Runic Swedish, from older *z* and with a rune of its own, the yr-rune ᛦ, stood among other places in endings such as *æftiʀ* 'after'; as it merged with r it was written ever more often with the r-rune.

Patrik Larsson (2002) mapped older ʀ in all Scandinavian runic inscriptions. After a vowel it is written with the r-rune markedly more often in Uppland (18 percent), Västergötland and on Gotland than in other provinces with many instances, and more often still in Västmanland (45), Närke (32) and Gästrikland (30 percent), which have few (table 4, p. 168; his measure counts spellings, ours stones). For Uppland, in his words, "knappast det geografiska läget i sig vara någon förklaring" (the geographical position in itself can hardly be an explanation); rather, "rimligen kronologiska, delvis kanske också individuella" (presumably chronological, partly perhaps also individual) factors have played a part (p. 170). The Uppland group in the Fp style is largely the carver Erik's, who consistently writes the r-rune for ʀ (section 7.2.3); Per Stille (1999, p. 206) inferred from the same that r and ʀ had merged in Erik's speech. Larsson sees the advance of the r-rune as a language change, not an orthographic reform (section 7.3): it has almost wholly taken over after dental consonants, less after other consonants and least after vowels, and the carvers mark palatalisation and write the s-rune for ʀ. Källström (2010) on individual carvers and the picture of language history and Källström (2014) on Erik we have not read.

What is lacking is a measure of whether the spelling clusters geographically more than chance would give, independently of provincial boundaries, and whether the cluster remains when the age of the stones and the habits of the carvers are held constant.

2. Pre-registration: what was fixed in advance and what is exploratory

No analysis plan was locked before the results were known. The finding was discovered in a systematic search of 13 traits, with the same method, limits and halves for all, fixed in the program code before the run and without any hypothesis about r for ʀ or western Uppland. Against the finding being an artefact of the choice stand the scan statistic's p-value, which takes every tested circle into account, the adjustment for the traits and for all 48 associations, the halves and the fixed sensitivity analyses. Sections 5.1–5.3 are what the method prescribes for every trait; 5.4 is exploratory, computed after the cluster was known, with unadjusted p-values.

3. Material

- **Source.** Rundata (2014 edition, updated 2018), retrieved 2026-10-04; derived corpus computed 2026-10-29. For one sensitivity analysis also the Swedish National Heritage Board's Runor (2020 edition), retrieved 2026-10-16.
- **The stones.** Viking-Age runestones (no other objects) in the Swedish provinces; Skåne and Blekinge are counted as Denmark in Rundata. Of the stones, 1,097 have a known original location and are not moved according to Rundata (72 of the 514 included are lost and known from older records). Stones that only Runor (2020) records as moved are included; they are left out in one of the sensitivity analyses.
- **The trait.** Words whose normalised form ends in ʀ and whose carved form ends in the r- or the yr-rune. The stone counts as *r for ʀ* if at least one such ending has the r-rune. The trait can be decided on 514 stones; 87 (17 percent) have it. The measure rests on Rundata's normalisation and transliteration, without a reading of our own.
- **The carver** according to Rundata, with a note of whether the attribution is secure (after Sveriges runinskrifter and Axelson 1993). Erik has twelve securely attributed stones where the trait can be decided and three uncertainly attributed ones (U 732, U 755, U 1151). His only signed stone, U 1165, is not counted, since a gap keeps the words from being paired; it has the r-rune where ʀ is expected (*-ftir*, *airikr*).
- **Dating** by style group (Gräslund 2006); in 5.4, RAK–Pr2 are early and Pr3–Pr5 late.

4. Method

4.1 The search in R (R 4.6.1), locations in SWEREF 99 TM: *Moran's I* (Moran 1950; 8 nearest neighbours) for whether the trait clusters at all; *Getis-Ord Gi** (Getis & Ord 1992) for whether it is more common among a stone's neighbours than in the country (Benjamini–Hochberg across the stones); and *Kulldorff's spatial scan statistic* (Kulldorff 1997), which tests every circle holding at most a tenth of the stones against the national share. The most deviant circle is the cluster; its p-value comes from 999 random redistributions and is adjusted with Benjamini–Hochberg across the 13 traits.

4.2 The halves. The material was divided once for all analyses into two halves, whole parishes dealt alternately within each province; the cluster is compared in each half with Fisher's exact test.

4.3 Sensitivity analyses. Logistic regression, within against outside the cluster: raw; adjusted for dating period; the carver as a random effect, so that a prolific carver's habits are not counted as the district's; the parish as a random effect; each half on its own; without stones that Runor 2020 records as moved. Without one province at a time and location as a spatial surface are not made: the cluster lies almost wholly within one province (35 of 36 stones), and the comparison is itself geographic. The choice of the cluster is tested by the scan statistic and the halves; in the adjustment for all 48 associations the larger of the adjusted model's p and the scan statistic's q across the 13 traits (0.013) is used. The assumptions are checked with simulated residuals (DHARMa; Hartig 2022).

4.4 Exploratory. Shares by carver and dating period, Fisher's exact test, unadjusted.

4.5 Deviations and earlier versions. No plan existed, so no deviations can be reported. A popular article (13 October 2026) rested on an older version of the material, before a correction of which stones count as moved, and had the cluster at Grop-Norrby with a radius of 22 kilometres; it was recomputed on 31 October 2026 and now has the same figures for the cluster, Erik, Balli and Åsmund as the report.

5. Results

5.1 The cluster. The trait clusters geographically: Moran's I = 0.18 (p < 0.001). The most extreme area has its centre at U 1172 at Holm in Vittinge parish and a radius of 30.4 kilometres. There 24 of 36 stones have r for ʀ (67 percent), against 13 percent of the stones in the rest of the country: 5.06 times as common. p = 0.001, the lowest possible; adjusted for the 13 traits q = 0.013, and no other trait's cluster passes the adjustment. The cluster's stones lie in 20 parishes, most in Simtuna (5) and in Vårfrukyrka, Tillinge, Nysätra, Fröslunda and Jumkil (3 each). Half A: 85 against 12 percent (odds ratio 41, p < 0.001); half B: 57 against 15 percent (odds ratio 7.3, p < 0.001).

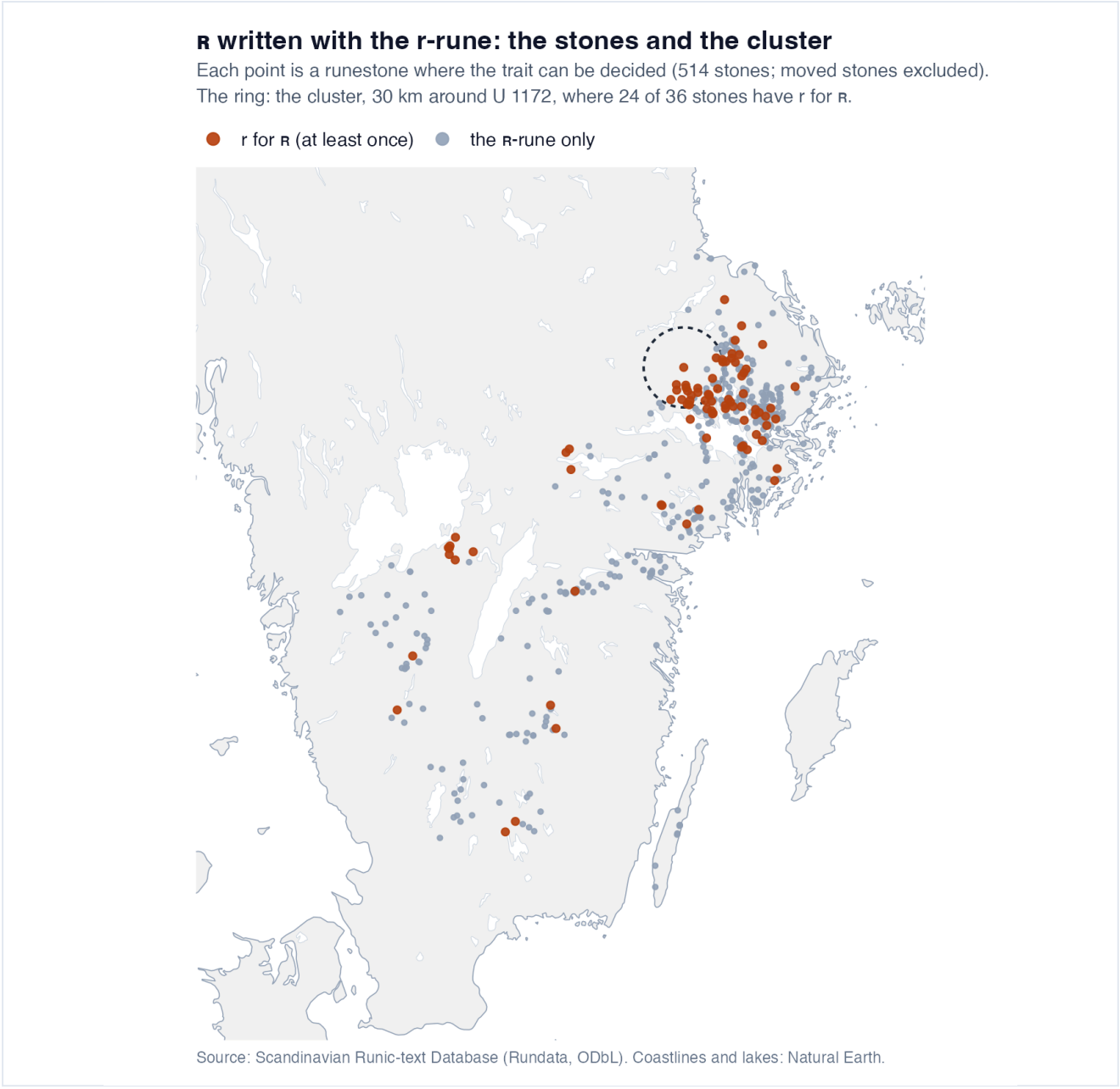


Figure 1. Runestones where the trait can be decided; rust: r for ʀ at least once, grey: the yr-rune only. The ring is the cluster. Map: Vitki, after Rundata (ODbL); coastlines and lakes: Natural Earth.

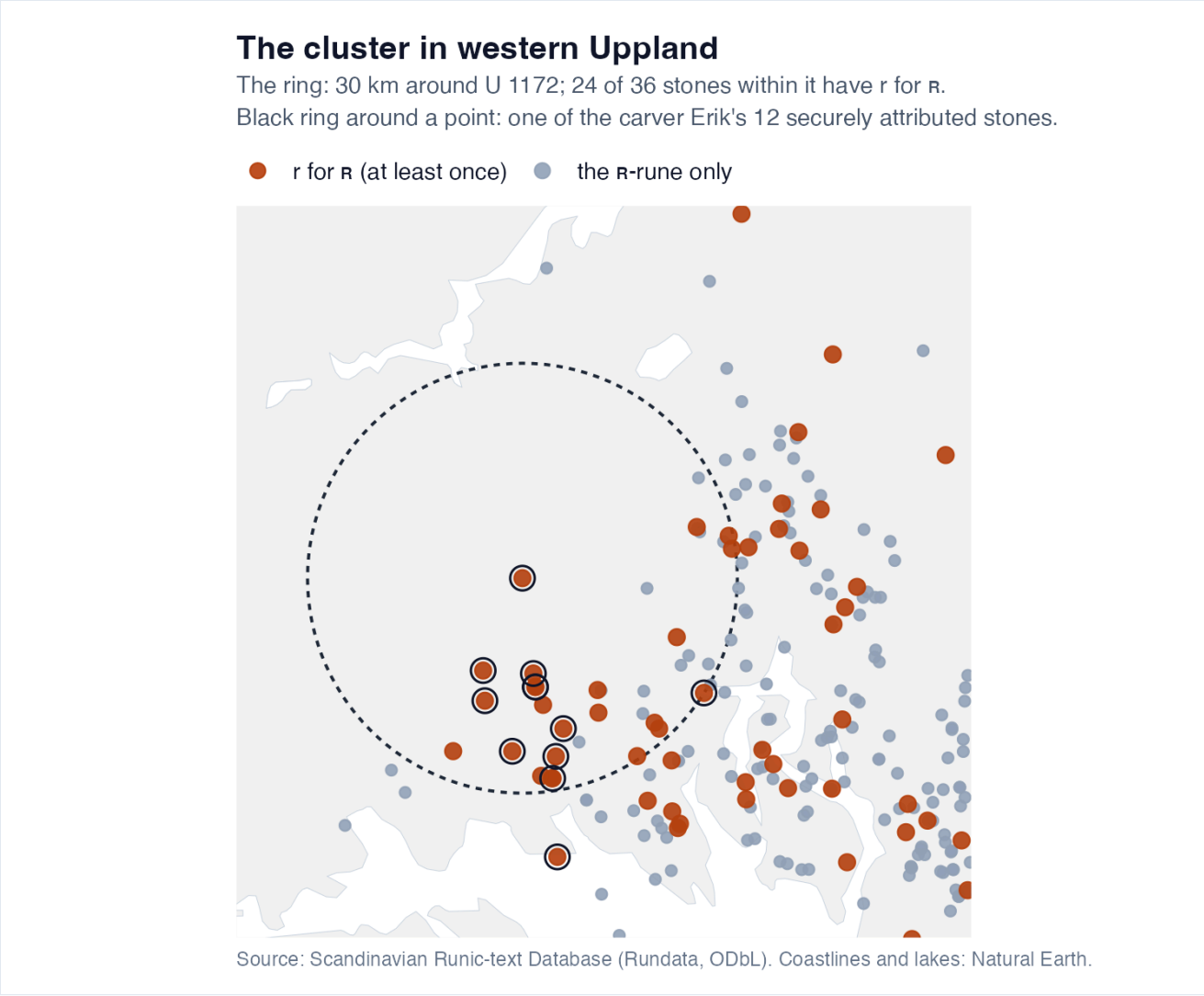


Figure 2. The cluster close up; black ring: one of Erik's securely attributed stones. Map: Vitki, after Rundata (ODbL); coastlines and lakes: Natural Earth.

5.2 The sensitivity analyses (514 stones, 87 with the trait). All 7 hold; odds ratio (95 % interval):

- Raw: 13.2 (6.3–27.7), $p < 0.001$. Adjusted for dating period: 13.0 (5.7–29.2), $p < 0.001$. Without stones moved according to Rundata: the same as the adjusted model, since they are already left out.
- Carver as a random effect (272 carvers and parish groups): 16.7 (5.6–49.8), $p < 0.001$. Parish as a random effect (280 parishes): 15.9 (6.4–39.3), $p < 0.001$.
- Half A (262 stones): 10.0; half B (252): 19.0; both $p < 0.001$.
- Without stones that Runor 2020 records as moved (447 remain): 10.5 (4.2–26.1), $p < 0.001$.
- In the adjusted model 63 against 13 percent, 50 percentage points (probably 32–67). The residuals: distribution $p = 0.10$, overdispersion $p = 0.08$, spatial pattern $p = 0.15$; nothing amiss.
- Adjusted for all 48 associations: $q = 0.02$, one of 15 that hold in every sensitivity analysis.

5.3 The hot stones. Gi∗ singles out 32 hot stones and no cold ones: 23 in and around the cluster in Uppland and Västmanland and 9 in Västergötland (Vg 2, Vg 3, Vg 4, Vg 6, Vg 12, Vg 13, Vg 15, Vg 16, Vg 106), a group too small to be tested as a cluster of its own. Larsson (2002, chapter 7) noted these in-

scriptions in Västergötland, where *æftiʀ* is written with the r-rune. The rest of Uppland: 39 of 244 (16 percent).

5.4 Exploratory: the carvers and time.

- **Erik.** All his 12 securely and three uncertainly attributed stones have r for ʀ, all in the Fp style; two stand outside the cluster, U 774 (secure, 39.7 kilometres from the centre) and U 732 (uncertain).
- **Early stones.** Within the cluster all 15 have the trait (13 attributed to Erik), outside 34 of 182 (19 percent).
- **Without Erik.** Without every stone attributed to him: 11 of 23 within the cluster (48 percent) against 63 of 478 outside (13 percent), $p < 0.001$.
- **Later stones** (Pr3–Pr5): 7 of 16 within the cluster (44 percent) against 12 of 109 in the rest of Uppland (11 percent), $p = 0.003$.
- **Balli**, a generation after Erik: 3 of 5 stones within the cluster against 5 of 19 outside ($p = 0.29$); with uncertain attributions 5 of 7 against 6 of 20 ($p = 0.08$).
- **Åsmund** wrote the yr-rune on all his 13 securely attributed stones, including the 3 within the cluster.
- **The attributions.** 33 of the cluster's 36 stones (92 percent) have a named carver, secure or uncertain, against 42 percent outside and 66 percent in the rest of Uppland; in Fjädrundaland all 45.

In plain words. Within thirty kilometres of Holm in Vittinge two stones in three have ʀ written with the ordinary r-rune at least once, in the rest of the country about one in eight, and in none of 999 random redistributions of the spelling did any area become as extreme.

6. Discussion

What the finding says. Western Uppland is the area of Sweden where ʀ was most often written with the r-rune, and that is not because of early stones or a single prolific carver: the cluster remains with dating period and carver constant, in each half, without moved stones and without Erik.

Against earlier research. The result sharpens Larsson's (2002) observation that Uppland and Västergötland stand out, recovers his group in Västergötland and confirms Stille (1999) on Erik. Larsson explains Uppland's share by time and individual carvers rather than by location; within Uppland there is nonetheless a bounded area that remains with both held constant. It borders on Västmanland, where Larsson has 45 percent (on 27 instances); perhaps this is a western zone rather than an Uppland trait.

Where we depart from earlier research. Stille and Larsson read the r-rune as pronunciation. Larsson's reasons concern runic writing at large: they show that many carvers wrote by ear, not that all did, and do not explain why one district differs so sharply. For western Uppland we propose a writing tradition that began with Erik and was passed on in the district:

1. Erik wrote r on all fifteen stones in the material securely or uncertainly attributed to him, including the two outside the cluster: U 774, 9 kilometres outside the circle, with a questioned attribution (Jansson in Upplands runinskrifter 3, pp. 324, 352, after Larsson 2002, section 7.2.3), and the uncertain U 732, 6 kilometres outside. Where Larsson, after Peterson (1983), sees the carvers' wavering between the runes as a sound change in progress, Erik does not waver; he writes as if by a rule.
2. Åsmund wrote the yr-rune on all his thirteen stones, including the three within the cluster; the place did not decide his spelling, but these are only three stones.
3. The spelling lives on in the district after Erik, and the cluster remains with the carver held constant.

With Åsmund the spelling followed the carver; Erik's stones all stand in or just outside the district, so for him, as for the rest of the district, the material cannot separate carver from place. The cluster remains with the carver held constant. If the carvers spoke a variety in which the sounds had merged, that gives the same pattern. Balli wrote r more often within the cluster than outside it, which fits both explanations, but on too few stones to tell from chance (5.4). The test is whether the later r-stones stand in Erik's tradition – ornament, formulas, spelling otherwise – or merely in his district.

Weaknesses, in order of weight.

1. **Discovered, not pre-registered.** The boundary is chosen by the data; radius and centre are approximate.
2. **Erik carries the early part.** 13 of 15 early stones in the cluster are attributed to him by style, and several attributions have been questioned (Larsson 2002, section 7.2.3); without him the margin is smaller.
3. **Uneven attributions.** The attributions are denser in the cluster (33 of 36 against 42 percent), largely because Stille (1999) attributed the stones of Fjädrundaland systematically (22 of the cluster's 36 stones lie there); the comparisons of carvers rest on uneven ground.
4. **Few words, no hand check.** Often one or two endings per stone, mostly *æftiʀ*, and a single r is enough; a misread yr-rune or a different normalisation moves individual stones.

What would overturn the finding. A hand reading in which the r-forms in the cluster are more often uncertain than outside it, or new finds in the district with the yr-rune.

7. Conclusion

Within 30.4 kilometres of Holm in Vittinge, 67 percent of the runestones have at least one r for ʀ, against 13 percent in the rest of the country, and the cluster withstands every sensitivity analysis. With Åsmund the spelling followed the carver; Erik's stones all stand in or just outside the district, so for him, as for the rest of the district, the material cannot separate carver from place. The cluster remains with the carver held constant. We propose a writing tradition that began with Erik; pronunciation, as Stille and Larsson hold, is not yet ruled out.

Data and code

The analysis was made in Vitki 3.0.0, an open research tool for runestones (vitkirunor.se). Code state: commit 4f80a63 in https://github.com/Precatio/runes (Vitki 3.0.0 plus later changes); the report's builder for the versions of 2026-11-05 from commit 6eae0af. The code and result files are there under GPL-3.0.

- The trait: `src/language_profile.py`, `src/pattern_map.py`; the search: `r/hotspots.R`; the sensitivity analyses: `r/robustness.R`, `src/robustness.py` (`METHODS.md`, 1f and 1v); the results under the material's version `e83202e01a5e6d27`; the figures: `utdata/rapporter/figurer/karta_r_for_R.R`. The report is built by `utdata/blogg/build/build_rapport_r_for_R.py`, which stops if any figure disagrees.
- In Vitki: Statistics, Geography, and Map, Patterns. Glossary: https://vitkirunor.se/en/forskningsrapporter/begrepp.
- Popular account: Erik wrote r – and the district followed, recomputed on 31 October 2026 with the same figures as here.

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Change log

- **Version 2, 2026-11-01.** The captions now say that the maps were made by Vitki. No numbers, interpretations or conclusions have changed.
- **Version 3, 2026-11-05.** Corrections before archiving in Zenodo: the filter for moved stones (only Rundata's moved stones are left out; 72 stones are lost), the number of sensitivity analyses (7, not 8), the claim that the spelling followed the carver rather than the place (holds for Erik and Åsmund), the title and the plain-words sentence (share of stones with at least one r for ʀ) and the status of the popular article. Added: Erik's stones outside the cluster, the density of attributions (Stille 1999), more of Larsson (2002), Peterson (1983), two unread works, code state, data versions, absolute links and sources in the captions. No analysis was redone and no earlier figure changed.
- **Version 4, 2026-11-05.** Corrections after a second review before archiving in Zenodo. The method of the adjustment for all 48 associations was described wrongly: it uses the larger of the adjusted model's p and the scan statistic's q across the 13 traits (0.013), not the halves' p (4.3). The list of the parishes with most stones lacked Nysätra, Fröslunda and Jumkil, which also have 3 stones each (5.1). That the spelling followed the carver holds only for Åsmund: Erik's stones all stand in or just outside the district, so for him carver and place cannot be told apart (abstract, discussion, conclusion and essentials box). Balli's figures are now said to be too few to tell from chance. The denser attributions in the cluster are explained largely, not wholly, by Stille's attributions in Fjädrundaland (22 of the cluster's 36 stones lie there). The code state now also names the commit holding the report's builder. Version 3 removed without saying so: that the later stones of the district write r four times as often as the rest of Uppland (in the abstract, discussion and conclusion; the figures remain in 5.4), that 7 of 9 Fp stones in Uppland outside the cluster have the trait, the shares in Södermanland (7 of 80) and Västergötland (9 of 41) and the reference to the method description. No analysis was redone and no earlier figure changed.

Competing interests: none.

Funding: none.

Cite as

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